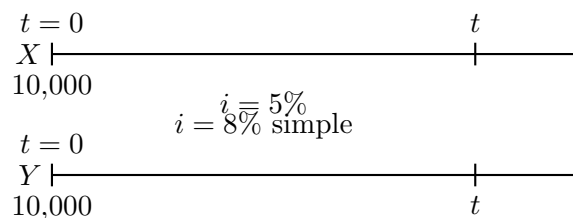


Problem 10.30

At time 0, deposits of 10,000 are made into each of Fund X and Fund Y . Fund X accumulates at an annual effective interest rate of 5%. Fund Y accumulates at a simple interest rate of 8%. At time t , the forces of interest on the two funds are equal. At time t , the accumulated value of Fund Y is greater than the accumulated value of Fund X by Z . Determine Z .

Solution

The timelines for the two funds are



For Fund X , the force of interest is constant:

$$\delta_X = \ln(1.05) = 0.048790.$$

For Fund Y , the accumulation function and force of interest are

$$a_Y(t) = 1 + 0.08t, \quad \delta_Y(t) = \frac{a'_Y(t)}{a_Y(t)} = \frac{0.08}{1 + 0.08t}.$$

Since the forces of interest are equal at time t ,

$$0.048790 = \frac{0.08}{1 + 0.08t},$$

$$t \approx 7.9959.$$

The accumulated values at that time are

$$A_X(t) = 10,000(1.05)^{7.9959} \approx 14,773.66,$$

$$A_Y(t) = 10,000(1 + 0.08(7.9959)) \approx 16,396.75.$$

Therefore,

$$\boxed{Z = A_Y(t) - A_X(t) \approx 1,623.09}.$$